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# Beyond Subject-Specific Models in Dynamical Human–Machine Interaction: Benchmarking and Optimization Strategies

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**Abstract**—Continuous finger position estimation from surface electromyography (EMG) enables smoother, more intuitive control in human–machine interfaces than discrete gesture classification. Accurate regression, however, requires effective temporal modeling and adaptation to user variability. We benchmark recurrent neural networks, temporal convolutional networks (TCNs), Transformers, and, for the first time in this context, neural ordinary differential equations (NODEs) on the Ninapro database. Each model’s receptive field (RF) is systematically tuned from EMG autocorrelation to ensure fair comparison. We further introduce a nonautonomous NODE variant with external inputs to represent finger movements as dynamical systems. To address cross-subject generalization, we explore adaptive learning paradigms, multitask, transfer, and first-order meta-learning, and adapt lightweight fine-tuning methods such as LoRA and adapter layers for temporal biosignal regression. On Ninapro DB8, the TCN achieves state-of-the-art mean absolute errors (MAEs) below 5.4 for multitask and transfer learning, and 6.47 for two-shot meta-learning. These findings advance EMG-to-kinematics regression and offer practical solutions for personalized, real-time control in prosthetics, virtual reality (VR), and teleoperation.

**Index Terms**—Human–machine interaction (HMI), meta-learning, neural-ODE, temporal convolutional networks (TCNs), transfer learning, Transformers.

## I. INTRODUCTION

**I**N HUMAN–MACHINE interaction (HMI), gesture recognition has traditionally enabled devices like prosthetic limbs to perform discrete movements from electromyography (EMG) signals [1], [2]. While effective for specific tasks, this approach limits expressiveness and fluidity, falling short of the continuous, natural motion required for intuitive and dexterous control.

Achieving seamless manual dexterity requires moving beyond gesture classification to continuous finger position

estimation, a regression approach that more faithfully captures natural hand motion and underpins diverse HMI applications. In prosthetics, it enables lifelike transitions between grips [3]; in virtual reality (VR), it enhances immersion through accurate motion tracking [4]; and in teleoperation, especially surgical robotics, it ensures precise, safe manipulation [5]. More broadly, human–machine interaction (HMI) spans augmented-reality teleoperation [6], impedance-controlled exoskeletons [7], and multimodal sensing [8], all of which require flexible, adaptive systems that handle complex user input—exactly what continuous EMG decoding targets. However, building such regression models is challenging. They must produce high-fidelity, low-latency outputs to follow subtle motion, and even small errors can degrade usability, making accurate, real-time EMG-to-kinematics mapping essential. These real-time demands are further complicated by generalization issues: signal characteristics vary across sessions due to electrode shifts and fatigue, while user-specific biomechanics alter the mapping between EMG and movement. Unlike classification, which yields comparable desired output distributions across users in controlled experimental settings, regression must adapt to these individual kinematic differences.

Regression-based EMG decoding is a relatively new area, with limited datasets supporting continuous estimation. Among them, Ninapro DB8 is widely adopted [9]. Unlike classification datasets, it enables continuous position estimation by mapping EMG signals to five degrees of actuation (DoA), derived from 18 degrees of freedom (DoF) that represent major hand movements.

Several deep learning models have been applied to this task, including temporal convolutional networks (TCNs) [10], Transformers [11], [12], and hybrid methods combining event-based encoding with linear regression [13]. TCNs currently offer a strong balance of accuracy and efficiency under real-time constraints [14]. Filters [9] and recurrent networks [15] have also contributed to early advances in this domain.

This article presents a comprehensive benchmark of temporal neural networks for EMG-based finger regression and introduces novel strategies to optimize both model performance and generalization. Although TCNs [10], [16], TEMPO-Net [17], long short-term memory (LSTM) networks, and Transformers [11], [12] have been used previously, we systematically optimize these architectures by tuning their memory to the autocorrelation structure of EMG signals,

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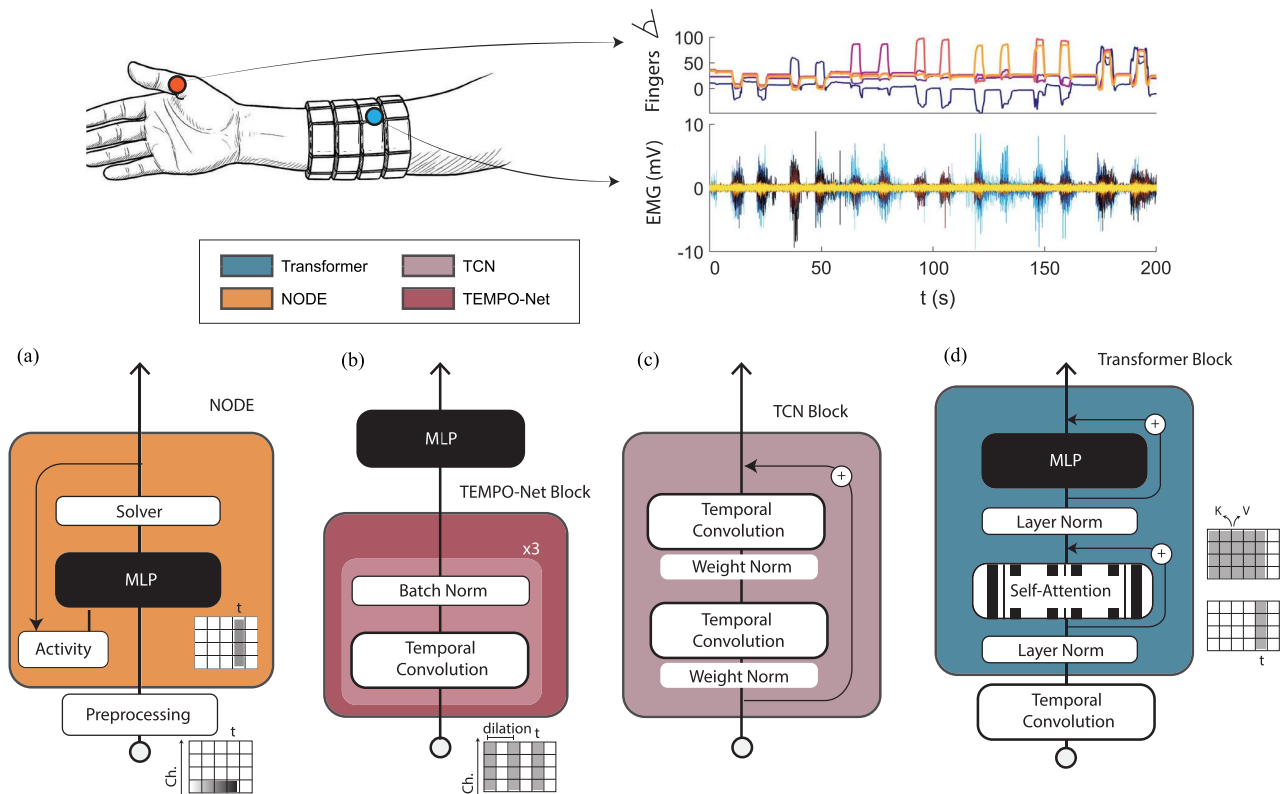


Fig. 1. Overview of the regression task to map EMG signals to finger movements. EMG signals from the forearm are mapped to finger angles using various neural network architectures, including NODE (a), TEMPO-Net (b), TCN (c), and Transformer (d).

improving accuracy under the same methodology for a fair comparison. We further introduce neural ordinary differential equations (NODEs) in this context.

Extending beyond model optimization and benchmarking, we tackle the challenge of cross-subject generalization. By treating each subject as a separate task, we use multitask learning, transfer learning, and first-order meta-learning to learn subject-specific models that can exploit the wider dataset and adapt with relatively few samples of each gesture.

Specifically, multitask learning [18] leverages shared and subject-specific parameters to enhance performance. Transfer learning enables pretraining on population data and fine-tuning for new users. Meta-learning enables rapid adaptation with minimal user data, a critical advantage in real-world scenarios where data acquisition is costly or time-constrained. To achieve this, the meta-learning literature include recurrent-based/in-context methods [19], [20], [21], learned optimizers that emit parameter updates [22], [23], [24], [25], [26], [27], and gradient-based initialization schemes [28], [29], [30]. While extensively studied for classification, these methods remain underexplored for regression, particularly in dynamic and noisy domains. Given the generalization challenges outlined above, we adopt gradient-based meta-learning, which adapts each model via gradient descent to improve cross-task transfer through a general-purpose optimization strategy.

To summarize, the key novelties of this work are: 1) a systematic benchmark of state-of-the-art and proposed models, with task-specific receptive field (RF) tuning; 2) the first application of NODE to EMG-to-kinematics regression; and

3) a comprehensive analysis of optimization and generalization strategies, including lightweight fine-tuning techniques such as LoRA, adapter layers, and a meta-learning TCN for few-shot temporal regression. Together, these advances establish a new benchmark for EMG-to-kinematics regression and provide practical paths toward scalable, user-adaptive HMI control in prosthetics, VR, and teleoperation.

## II. MATERIALS AND METHODS

The methodology is structured in three parts. First, we describe the five benchmarked architectures for continuous finger-movement regression from forearm surface EMG signals: LSTM, TCN, TEMPO-Net, Transformer, and NODE (see Fig. 1). Second, we outline the optimization and fine-tuning strategies, including multitask, transfer, and meta-learning, as well as LoRA and adapter layers. Finally, we detail the dataset and evaluation metrics, with full specifications provided in the Supplementary Material.

### A. Network Architectures

1) *Recurrent Neural Networks*: LSTM networks [31] are foundational recurrent neural network (RNN) architectures widely used in neuroprosthetics [15]. They effectively capture long-range temporal dependencies while mitigating vanishing gradients. Like other recurrent models, LSTMs scale linearly with sequence length but compress temporal context into fixed-dimensional hidden states, limiting their memory capacity. Recent advances have sought to bridge this gap

with transformer-inspired recurrent variants [32], [33], [34]. Here, we employ a standard LSTM as a baseline to evaluate the suitability of traditional recurrent architectures for EMG-to-kinematics regression.

2) *Temporal Convolutional Networks*: TCNs apply 1-D convolutions over time, enforcing temporal translation equivariance and capturing local temporal dependencies through joint processing of neighboring samples. We benchmarked two variants: TEMPO-Net, designed for EMG data, comprising three convolutional blocks with varying strides and batch normalization followed by a multilayer perceptron (MLP) that concatenates layer responses as in LeNet5 [35] [see Fig. 1(b)]; and a residual TCN employing causal dilated convolutions and weight normalization [36], which preserves sequence length and outputs via a point-wise projection [see Fig. 1(c)].

3) *Transformers*: Despite the quadratic scaling of standard self-attention, we include a lightweight Transformer in our benchmark to reflect its growing use across domains, including EMG, and the generality of the self-attention mechanism. This enables a broader evaluation of expressiveness versus efficiency in EMG-to-kinematics mapping. The Transformer architecture used [see Fig. 1(d)] is analogous to the one proposed in [11] and [12], where the signal is first encoded by a convolutional layer and then processed by multiple blocks of windowed self-attention [12]. Windowed self-attention restricts computation of keys and values to a temporal window  $[t - W\delta t, t]$ , while the query accesses only the current time step, keeping computational load low for real-time use. Unlike prior EMG work, we employ sinusoidal positional encodings [37] with a minimum frequency matched to the attention window, extending the RF through strides and DS while reducing overfitting at longer timescales.

4) *Neural Ordinary Differential Equations*: A NODE models temporal dynamics as an ordinary differential equation in which a neural network learns the derivative  $\dot{\mathbf{z}}(t)$  rather than predicting  $\mathbf{z}(t)$  directly. The system's evolution is obtained through numerical integration, introducing an inductive bias toward continuous-time representations. The proposed nonautonomous NODE is defined as

$$\frac{d\mathbf{z}(t)}{dt} = \mathbf{f}(\mathbf{z}(t), \mathbf{g}(\mathbf{x})|\theta) \quad (1)$$

$$\mathbf{z}(t_0) = \mathbf{z}_0 \approx \mathbf{g}_0(\mathbf{x}) \quad (2)$$

where  $\mathbf{f}$  is an MLP parameterized by  $\theta$  and  $\mathbf{g}(\mathbf{x})$  represents the driving signal [see Fig. 1(a)].

In our typical NODE model,  $\mathbf{g}(\mathbf{x})$  is a preprocessing routine, though we also explored trainable neural network transformations. This transformed input is essential because standard neural ODEs, parameterized by static MLPs, lack memory and operate under a Markovian assumption. Since the task relies on capturing temporal dependencies, we embed memory into  $\mathbf{g}(\mathbf{x})$  through preprocessing. Instead of learning  $\mathbf{g}(\mathbf{x})$  via a neural network, we follow prior work [13], [38] by applying Kalman filters and convolving the input with an exponential decay kernel (timescale  $\approx 0.1$  s), efficiently capturing some temporal context with minimal computational overhead.

The model in (1) requires initial conditions  $\mathbf{z}(t_0)$ . We set  $\mathbf{z}(t) \equiv \hat{\mathbf{y}}(t)$ , treating directly finger angles as the dynamical

system to be modeled. This allows straightforward training by initializing with ground truth ( $\mathbf{z}(t_0) = \mathbf{y}(t_0)$ ) and predicting the next  $L$  time steps. At inference, we test two modes: initializing with a single test value and forecasting future steps; estimating  $\mathbf{z}(t_0)$  using a separate network  $\mathbf{g}_0(\mathbf{x}|\theta_0)$  with periodic resets. The latter mitigates error accumulation typical in long autoregressive sequences. For  $\mathbf{g}_0(\mathbf{x}|\theta_0)$ , we employed a TCN, which is well suited to provide past delayed inputs [39], [40] and estimate higher order derivatives. In practice, both inference modes performed similarly on the task considered, as long as the reinitialization in the second mode occurred at multisecond intervals, allowing the model to rely primarily on (1) rather than (2). Finally, we explored neural ODEs to assess the benefits of a continuous-time sequential model for this task, noting that such models offer greater expressivity than alternatives like classical Kalman filters (see Supplementary Material).

## B. Network Optimization Approaches

Given  $N_{\text{subj}}$  subjects and the temporal nature of the task, the dataset is formalized as a collection  $\mathcal{D} = \{D_i\}_{i=1}^{N_{\text{subj}}}$ , where each  $D_i = \{(\mathbf{x}(t), \mathbf{y}(t))\}_{t=1}^T$  represents subject-specific data comprising  $T$  time-indexed input-output pairs.

1) *Subject-Specific and General Models*: Following prior work, our first goal is to learn subject-specific models  $\mathbf{f}(\cdot|\theta_i^*)$  by minimizing:

$$\theta_i^* = \arg \min_{\theta_i} \mathcal{L}_i(\mathbf{f}(\cdot|\theta_i)) \quad (3)$$

where  $\mathcal{L}_i$  denotes the loss computed on  $D_i$  and  $\theta_i$  are the model parameters specific to subject  $i$ .

After benchmarking subject-specific systems, the next natural step is to generalize across individuals by training on multiple subjects. This leads to the optimization problem

$$\theta^* = \arg \min_{\theta} \sum_i \mathcal{L}_i(\mathbf{f}(\cdot|\theta)) \quad (4)$$

where all subjects are treated with equal weight. In this formulation, the model must identify a parameter set that captures patterns shared across individuals. The resulting performance will naturally depend on the degree of intersubject variability and the overlap among individual training distributions.

2) *Multitask and Transfer-Learning*: Multitask learning exploits shared structure across tasks by jointly optimizing task-agnostic  $\theta$  and task-specific parameters  $\delta\theta_i$ . This formulation aims at improved performance, assuming that tasks benefit from common representations. In our simulations, this corresponds to a two-stage training procedure: first, a shared model is trained across all subjects to learn  $\theta^*$  via (4), followed by subject-specific fine-tuning to obtain  $\delta\theta_i^*$  solving:

$$\delta\theta_i^* = \arg \min_{\delta\theta_i} \mathcal{L}_i(\mathbf{f}(\cdot|\theta^* \cup \delta\theta_i)). \quad (5)$$

While this process can offer performance gains, it requires simultaneous access to all subjects' data, an assumption that limits real-world applicability. For deployment, it is preferable to fine-tune models on unseen users without relying on preexisting source datasets. This scenario is naturally cast as



a transfer learning problem. To evaluate performance under these conditions, we adopted a leave-one-subject-out cross-validation strategy. We first pretrain the model on a reduced dataset  $\mathcal{D}^{-i} = \mathcal{D} - D^i$ , excluding data from subject  $i$ , and subsequently fine-tune the model on  $D^i$

$$\theta_{-i}^* = \arg \min_{\theta} \sum_{j \neq i} \mathcal{L}_j(\mathbf{f}(\cdot | \theta)) \quad (6)$$

$$\delta \theta_i^* = \arg \min_{\delta \theta_i} \mathcal{L}_i(\mathbf{f}(\cdot | \theta_{-i}^* \cup \delta \theta_i)) \quad (7)$$

Performance is then assessed on the test portion of  $D^i$ . This procedure is repeated across all subjects  $i = 1, \dots, N_{\text{subj}}$ , yielding  $N_{\text{subj}}$  distinct models, each trained on  $\mathcal{D}^{-i}$ , fine-tuned on  $D^i$ , and evaluated on its respective test subset.

3) *Meta-Learning*: Meta-learning [41], particularly in its gradient-based form [28] and [29], aims to optimize the initial parameter configuration  $\theta$  to enable rapid adaptation to new tasks using only a few data points. This is achieved through a two-timescale optimization process involving an inner and an outer (meta) loop. The inner loop performs fast adaptation to a task using task-specific data, while the outer loop gradually updates parameters based on the adaptation trajectories. This hierarchical structure allows the model to acquire higher-order knowledge about the optimization landscape. Increasing the number of inner-loop updates enhances expressiveness but also deviates from standard training procedures and increases computational cost.

In this work, we focus on first-order meta-learning methods due to their scalability and computational efficiency. A full derivation from first principles is provided in the Supplementary Material; here, we present the general formulation of gradient-based first-order algorithms. Let Alg denote a generic inner-loop update rule applied over  $M$  steps. The bi-level optimization can be expressed as

$$\theta_{i,m} = \text{Alg}(\hat{\mathcal{L}}_{i,m}(\theta_{i,m-1}), \dots, \hat{\mathcal{L}}_{i,1}(\theta)), \quad m = 1, \dots, M \quad (8)$$

$$\frac{\partial \mathcal{L}}{\partial \theta} \underset{\text{F.O.}}{\approx} \sum_i \sum_{m=1}^M \alpha_m \frac{\partial \mathcal{L}_{i,m}(\theta_{i,m})}{\partial \theta_{i,m}} = \sum_{m=1}^M \alpha_m g_m. \quad (9)$$

Here, the hat notation ( $\hat{\mathcal{L}}$ ) indicates inner-loop losses, while  $\theta$  represents the initial parameters before adaptation. The meta-objective in (9) is computed using a first-order approximation that neglects second-order derivatives (Hessian terms). The index  $i$  refers to individual tasks and  $m$  to the step within the inner loop. The weights  $\alpha_m \in [0, 1]$  modulate the contribution of each step to the overall meta-gradient, denoted  $g_m$ . In this description, the variants of FOMAML and Reptile [28] differ only in the choice of support set  $\hat{\mathcal{L}}$ , query set  $\mathcal{L}$ , and the coefficients  $\alpha_m$ ; full details are provided in the Supplementary Material. For the regression task considered, adopting a meta-learning algorithm poses two main challenges: the temporal nature of the problem complicates data handling, specifically, the choice and alignment of the input–target segments used to compute the losses  $\mathcal{L}_{i,m}$ ; the tasks are not mutually exclusive, making task boundaries less distinct [42]. To address these, we implemented data augmentation to define distinct mini-batches within the inner loop and used the Adam optimizer for both inner and meta-level updates. This sampling strategy

ensures that each inner-loop batch differs, enabling gradient reuse similar to Reptile. The main distinction between first-order MAML (FOMAML) and Reptile in this scenario lies exclusively in the choice of  $\alpha_m$ , that is, the weighting of inner-loop steps (see Supplementary Information). FOMAML adopts a backward perspective by prioritizing later steps, while Reptile emphasizes a forward view, considering the full trajectory. In the Results section, we compare different  $\alpha_m$  schedules and assess their effect on performance. For training and evaluation, we adopted the same data-splitting strategy as in the transfer learning setting: a single  $D^i$  was held out for few-shot evaluation, and the meta-learning procedure was repeated across all subjects varying  $D^i$ .

4) *Efficient Fine-Tuning Techniques*: During multitask learning and transfer learning, we explored low-rank adaptation (LoRA) and Adapter layers for fine-tuning of the additional parameters  $\delta \theta_i$ , imposing that this adaptation process occurs in a lower-dimensional space in comparison to the pretraining phase. LoRA decomposes the weights' changes as the product of two low-rank matrices. For a TCN and an example convolutional layer with weights  $\theta_{jkl} \in \mathcal{R}^{K \times C_{\text{out}} \times C_{\text{in}}}$ , where  $K$ ,  $C_{\text{out}}$ , and  $C_{\text{in}}$  are the kernel size, number of input and output channels, respectively, we adopted the following LoRA decomposition:

$$\delta \theta_{jkl} = \sum_n^{\text{rank}} A_{jkn} B_{ln} \quad (10)$$

where  $\delta$  refers to the parameter variations due to fine-tuning, and the rank value governs the dimensionality of the adaptation process.

Adapter layers offer an alternative fine-tuning strategy, originally introduced in the context of large language models (LLMs). They consist of additional low-dimensional layers with residual connections inserted into each block of the network. Like LoRA, adapters constrain adaptation to a low-dimensional subspace; however, unlike LoRA, the inclusion of full layers introduces nonlinearity. We included two variants of adapter layers for fine-tuning. In the first, we followed the original design used in LLMs, implementing coordinate-wise adapter layers within each block of the networks considered. In the second variant, we explored adapters based on temporal convolutions to enable stronger mixing of temporal features during the fine-tuning stage, similar to LoRA. Specifically, we used temporal convolutions with a small number of output channels to define a low-dimensional space, followed by a second convolution that projects back to the original channel dimensions.

## C. Dataset and Evaluation Metrics

1) *Data for the Regression Task and Data Processing*: This study utilizes the existing Ninapro DB8 dataset, which comprises ten non-amputee subjects and two transradial amputee subjects [9]. Participants performed a set of nine mirrored bilateral hand movements, captured via 16 active double-differential EMG sensors (Delsys Trigno IM Wireless EMG system) positioned around the right forearm. Movements were repeated multiple times across three different sessions.

Additionally, a Cyberglove 2 was used to record hand kinematics on the contralateral (left) hand, capturing 18DoF at 100 Hz, later upsampled to 2 kHz to synchronize with the EMG data. The experimental protocol was designed for smooth transitions between movements, each lasting 6–9 s, followed by a 3-s rest. The gestures include: thumb rotation, thumb flexion, index flexion, middle flexion, and combined ring-little flexion. In all our experiments, Sessions 1 and 2 were used for training and fine-tuning, while Session 3 was used consistently for testing to assess generalization. We selected the Ninapro DB8 dataset precisely because it is one of the very few publicly available resources containing both high-density EMG and glove-based continuous finger kinematics in non-amputee and amputee populations. Its widespread use in previous work makes it particularly well-suited for benchmarking and comparative analysis, allowing our results to be meaningfully positioned relative to the existing literature.

2) *Evaluation Metrics*: To evaluate performance, we utilized metrics such as mean absolute error (MAE), the RF, and principal component analysis (PCA)-based clustering. MAE served as the primary metric to quantify the prediction accuracy of continuous finger movements from EMG data, defined as

$$\text{MAE} = \frac{1}{N_{\text{infer}} N_{\text{DoA}}} \sum_{i=1}^{N_{\text{infer}}} \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|_1 \quad (11)$$

where  $\mathbf{y}_i, \hat{\mathbf{y}}_i \in \mathbb{R}^5$  are the multivariate ground truth and estimation, measured in degrees, corresponding to the  $i$ th inference, respectively,  $\|\cdot\|_1$  is the  $L_1$ -norm,  $N_{\text{DoA}} = 5$  is the number of DoAs, and  $N_{\text{infer}}$  is the total number of inferences in each subject's session. To evaluate each network's memory capabilities, we analytically computed the receptive field (RF) (see Supplementary Information), which denotes the temporal window over which the model can extract dependencies from the EMG input. Excluding models such as NODE and LSTM, where memory is embedded in internal dynamics, other architectures possess an explicit RF defined as the number of input time steps that influence the output at time  $t$ . For architectures with  $N$  residual connections, the network comprises multiple parallel paths, resulting in  $2^N$  possible RF values. In this case, we report the maximum RF, corresponding to the longest path.

A larger RF enables the network to capture long-range temporal dependencies, which are crucial for modeling continuous movements over extended durations. Although often overlooked, computing the RF for a given architecture is essential for interpretability and fair comparison. We report RF both in discrete time steps (i.e., the number of input samples that contribute to a prediction) and in seconds. For downsampled inputs, the temporal RF reflects the full duration spanned by the contributing samples, irrespective of intermediate gaps, emphasizing the effective time range the model can integrate over.

Finally, to visually assess model performance, we applied  $k$ -means clustering [43] and PCA to project the output space into two dimensions and identify movement clusters. This analysis was performed on predicted and target trajectories. The overlap and separation of clusters provided insights into

the model's ability to preserve temporal coherence and distinguish individual movement patterns from EMG signals.

3) *DS for Data Augmentation and Temporal Modeling*: To enhance model performance and generalization, the input EMG signals were sparsified in time by a downsampling (DS) factor. While EMG exhibits rapid stochastic fluctuations, the target kinematics evolve more slowly, with meaningful changes occurring over hundreds of milliseconds (as illustrated by the finger angle trajectories in Fig. 1). All models are trained in temporal segments with randomized starting points, with each minibatch comprising distinct segments. DS is applied after selecting the starting time, serving as a data augmentation strategy that introduces input variability and enhances generalization. Given that the models map fast-varying inputs to slower outputs  $\mathbf{y}(t)$ , DS by a factor of DS effectively yields DS distinct input sequences. These range from  $\mathbf{x}(t_j)$  to  $\mathbf{x}(t_j + \text{DS}\delta t)$  and correspond to nearly identical output values  $\mathbf{y}(t), \dots, \mathbf{y}(t + \text{DS}\delta t)$ . Each downsampled sequence contains unique input data, allowing the model to exploit output redundancy while potentially discarding fine-grained input details. This also extends the effective temporal window available for prediction and biases the model over longer timescales.

### III. RESULTS

#### A. Subject-Specific Model Performance

We first focus on subject-specific models, in line with previous works. Fig. 2(a) shows the performance of different models as the DS factor varies. As described in the Methods section, DS encourages the systems to capture statistics over longer timescales. This initially reduces MAE for all systems but also introduces the risk of overfitting when models prioritize long-term features that are not generalizable. All systems achieve a minimum MAE when the resulting RF is on the order of seconds, which matches the typical duration of a gesture and enables effective information integration. Notably, architectures with an explicit RF exhibit similar performance trends with approximately the same minimum. However, we emphasize that the inclusion of positional encoding was essential for the Transformer; without it, the model did not benefit from DS. This finding is interpretable in the sense that, at lower timescales, the temporal order of input information becomes increasingly important, making positional encoding crucial for the Transformer architecture.

Fig. 2(b) illustrates the evolution of the MAE over training iterations, averaged across all subjects and five runs, thereby capturing the model's learning dynamics and convergence behavior. The system configurations used for these results correspond to the DS conditions marked by the stars in Fig. 2(a).

The proposed TCN is reaching lower MAE values compared to the other models while featuring only  $6 \times 10^4$  trainable parameters and a number of multiply and accumulate operations  $\approx 1.9 \times 10^6$  for an example batch size of 32 (see Supplementary Information). The inset in Fig. 2(b) displays the training MAE, which is substantially lower than the test MAE, underscoring the challenge of generalizing across sessions with varying input–output statistics due to experimental

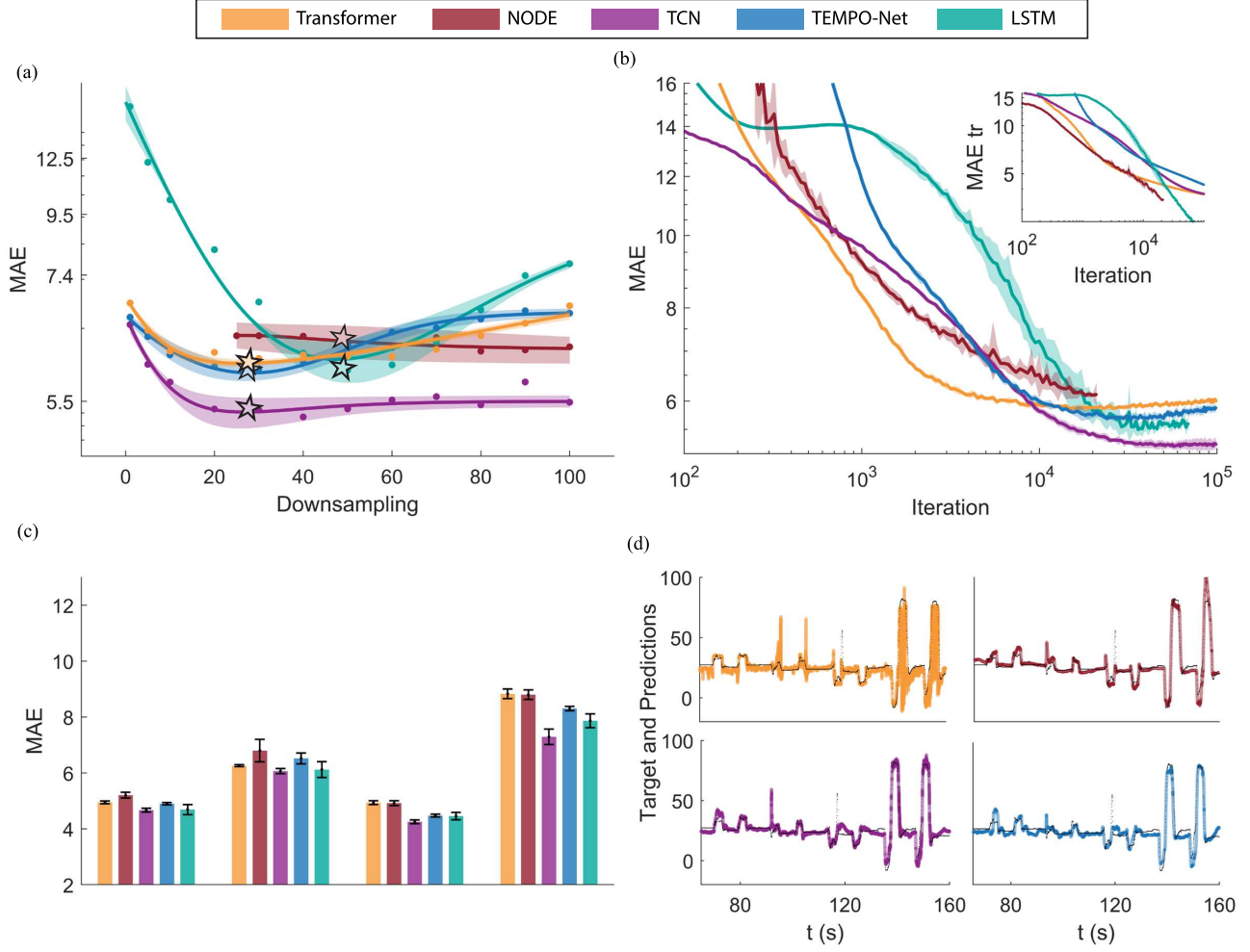


Fig. 2. (a) MAE trends as the DS factor varies, illustrating how the performance of different models changes as the systems are biased toward longer timescales and increased RF. (b) MAE trends over training iterations for models trained separately per subject; results are averaged across subjects to report performance, highlighting how effectively the model adapts to individual subject data, with TCN achieving lower MAE values more efficiently than the other systems. (c) MAE values for two non-amputee and two amputee subjects (one accident-related, one cancer-related) across each network. (d) Target and prediction view for each network, displaying one DoF for clarity. The stars indicate the DS values selected for use in subsequent simulations, including those shown in (b).

variability. Except for the Transformer and Tempo-Net, which exhibit minor performance degradation during training, the models do not show signs of over-fitting. The performance gap instead reflects the broader issue of intersession generalization, further addressed in the transfer learning results.

Fig. 2(d) presents a representative comparison between target and prediction outputs for each network, displaying a single DoF for clarity. The NODE model produces smoother output trajectories, consistent with its inherent bias toward modeling continuous dynamical systems. This property is beneficial in contexts where target signals exhibit gradual transitions. Fig. 2(c) reports the MAE for two non-amputee and two amputee subjects (one due to an accident, the other due to cancer) across all networks. The observed variability emphasizes subject-specific characteristics and indicates that specific model performance may differ across individuals.

### B. Generalized Model Performance Across Subjects

The cross-subject setting poses a greater challenge, as the model must learn features that generalize across individuals.

Performance in this scenario reflects the extent of behavioral overlap between users and provides insights into whether the task exhibits shared structure or is predominantly subject-specific.

Fig. 3 shows the performance of six models: two Transformers (with one and two blocks), a NODE, a TCN, an LSTM, and TEMPO-Net, trained on the combined dataset from all subjects. The LSTM and TCN achieve the lowest MAE, with the LSTM converging more rapidly. However, the TCN is significantly more efficient, requiring less than one-third of the trainable parameters and MAC operations (see Table I).

Among the remaining models, TEMPO-Net yields performance comparable to the TCN, though at a higher computational cost. The two-block Transformer demonstrates improved performance over the single-block version, highlighting the advantage of deeper attention mechanisms in modeling long-range temporal dependencies (with an RF of 1500 steps, or 0.7 s). The TCN also shows minimal performance variability across subjects, as indicated by the consistently low standard deviation in MAE, suggesting robust

TABLE I

PERFORMANCE COMPARISON OF ALL NETWORKS UNDER SUBJECT-SPECIFIC TRAINING, BASED ON MULTIPLE METRICS: MAE, MAE VARIATION, RF, AND NUMBER OF PARAMETERS. THE TABLE IS DIVIDED INTO PRIOR RESULTS AND SECTIONS CORRESPONDING TO THE DIFFERENT OPTIMIZATION STRATEGIES EXPLORED IN THIS STUDY. THE SUPERScript <sup>1</sup> ON OUR TEMPONET IMPLEMENTATION DENOTES THE USE OF THE ORIGINAL MODEL [17], WHICH INCLUDES MORE TRAINABLE PARAMETERS BUT SHOWS IMPROVED PERFORMANCE OVER THE UPDATED VERSION [10]. VALUES IN PARENTHESES FOR TRAINABLE PARAMETERS AND MMAC INDICATE ADDITIONAL CONTRIBUTIONS FROM FINE-TUNING MECHANISMS AND THEIR ASSOCIATED INCREASE IN OPERATIONS, RESPECTIVELY. MMAC WAS COMPUTED FOR A BATCH SIZE OF 32 FOR CONSISTENCY WITH LITERATURE BENCHMARKS. LoRA DOES NOT INCREASE MMAC, AS ITS MATRIX DECOMPOSITION INTRODUCES NO ADDITIONAL NONLINEARITIES AND CAN BE MERGED INTO THE MODEL PARAMETERS AT INFERENCE. FOR THE TRANSFER LEARNING RESULTS IN FEW-SHOT SETTINGS, THE VALUES REFER TO THE FULLY CONVERGED MODELS

| Model                                    | MAE         | $\sigma(\text{MAE})$ | Parameters                                   | MMAC       | RF                     |
|------------------------------------------|-------------|----------------------|----------------------------------------------|------------|------------------------|
| <b>Subject-specific (previous works)</b> |             |                      |                                              |            |                        |
| LSTM [15]                                | 7.04        | —                    | —                                            | —          | NA                     |
| SVM [44]                                 | 7.28        | —                    | —                                            | —          | NA                     |
| MLP [44]                                 | 7.14        | —                    | —                                            | —          | NA                     |
| Online-Transformer [12]                  | 6.10        | —                    | $\approx 9 \times 10^4$                      | 5.3        | $\approx 750$ (0.375s) |
| TempoNet [44]                            | 6.89        | —                    | $\approx 1.0 \times 10^5$                    | 3.2        | $\approx 500$ (0.250s) |
| <b>Subject-specific</b>                  |             |                      |                                              |            |                        |
| TempoNet <sup>1</sup>                    | 5.80        | <b>0.99</b>          | $\approx 3.0 \times 10^5$                    | 9.2        | $\approx 500$ (6.250s) |
| Online-Transformer                       | 5.91        | 1.10                 | $\approx 9 \times 10^4$                      | 5.3        | $\approx 750$ (1.875s) |
| TCN                                      | <b>5.45</b> | 1.06                 | $\approx 6 \times 10^4$                      | 1.8        | $\approx 475$ (6s)     |
| LSTM                                     | 5.61        | 1.16                 | $\approx 2.1 \times 10^5$                    | 6.8        | NA                     |
| Neural-ODE                               | 6.09        | 1.16                 | $\approx 5 \times 10^4$                      | <b>1.7</b> | NA                     |
| <b>Across-subjects</b>                   |             |                      |                                              |            |                        |
| TempoNet <sup>1</sup>                    | 5.78        | 0.89                 | $\approx 3.0 \times 10^5$                    | 9.2        | $\approx 500$ (6.250s) |
| Online-Transformer                       | 7.10        | 1.02                 | $\approx 9 \times 10^4$                      | 5.3        | $\approx 750$ (1.875s) |
| Online-Transformer (2 blocks)            | 6.01        | 0.95                 | $\approx 1.8 \times 10^5$                    | 10.3       | $\approx 1500$ (3.75s) |
| TCN                                      | 5.70        | <b>0.73</b>          | $\approx 6 \times 10^4$                      | 1.8        | $\approx 475$ (6s)     |
| LSTM                                     | <b>5.57</b> | 0.90                 | $\approx 2.1 \times 10^5$                    | 6.8        | NA                     |
| Neural-ODE                               | 6.86        | 1.30                 | $\approx 5 \times 10^4$                      | <b>1.7</b> | NA                     |
| <b>Multi-task Learning</b>               |             |                      |                                              |            |                        |
| TCN + LoRA                               | <b>5.33</b> | 1.00                 | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | <b>1.8</b> | $\approx 475$ (6s)     |
| TCN + Adapter-Layers                     | 5.40        | <b>0.98</b>          | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | 1.8 (+1.1) | $\approx 475$ (6s)     |
| TCN + Adapter-Layers with Conv           | 5.52        | 1.02                 | $\approx 6 \times 10^4$ (+ $2 \times 10^4$ ) | 1.8 (+1.0) | $\approx 475$ (6s)     |
| <b>Transfer learning</b>                 |             |                      |                                              |            |                        |
| TCN + LoRA                               | <b>5.39</b> | <b>1.03</b>          | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | <b>1.8</b> | $\approx 475$ (6s)     |
| TCN + Adapter-Layers                     | 5.55        | 1.13                 | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | 1.8 (+1.1) | $\approx 475$ (6s)     |
| TCN + Adapter-Layers with Conv           | 5.60        | 1.21                 | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | 1.8 (+1.1) | $\approx 475$ (6s)     |
| <b>Few-shot</b>                          |             |                      |                                              |            |                        |
| TCN (1-shot)                             | 8.70        | <b>1.18</b>          | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | 1.8        | $\approx 475$ (6s)     |
| TCN (2-shot)                             | 6.81        | 1.12                 | $\approx 6 \times 10^4$ (+ $3 \times 10^4$ ) | 1.8        | $\approx 475$ (6s)     |
| TCN + Meta-learning (1-shot)             | <b>8.22</b> | 1.21                 | $\approx 6 \times 10^4$                      | 1.8        | $\approx 475$ (6s)     |
| TCN + Meta-learning (2-shot)             | <b>6.47</b> | <b>1.01</b>          | $\approx 6 \times 10^4$                      | 1.8        | $\approx 475$ (6s)     |

and stable behavior. In contrast, the autoregressive NODE and the recurrent LSTM exhibit greater variability and more pronounced oscillations during training, reflecting their increased sensitivity to initialization and temporal noise.

### C. TCN Performance and Temporal Structure

In this study, we developed and refined several models, achieving improved performance over previous work (see Table I). These improvements are primarily attributed to the adopted sampling strategy, wherein temporal segments within each mini-batch are randomly initialized and downsampled. Other hyperparameter tuning focused on optimizing learning

rate and dropout values (see Supplementary Information). Based on the results in Figs. 2 and 3, and considering model efficiency relative to parameter count, the proposed TCN was selected for benchmarking optimization strategies. With  $\approx 5 \times 10^4$  trainable parameters, the TCN achieved a MAE of 5.45 under subject-specific training and 5.70 in the cross-subject setting (see Table I).

Fig. 4 summarizes the TCN model's performance and further emphasizes the importance of RF tuning. Fig. 4(a) illustrates how increasing the DS rate, thereby expanding the RF, enhances model performance, providing a focused view of a  $\sim 5$ –6-s window (DS from 5 to 50) from Fig. 2(b). As the DS rate increases, the model's RF expands



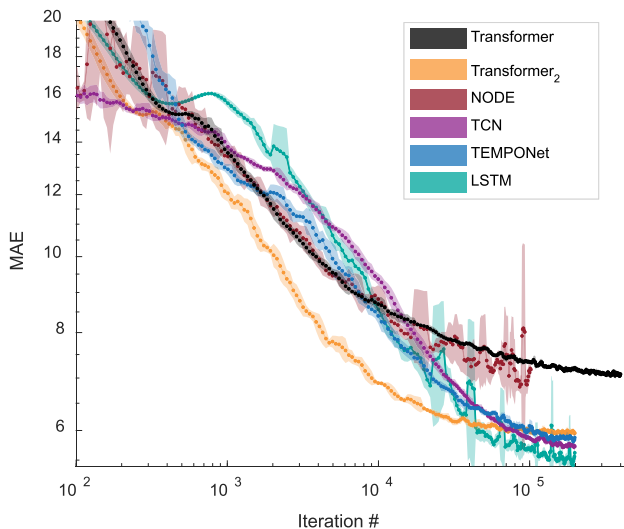


Fig. 3. Performance of multiple models trained on a combined dataset for all subjects: two Transformers (one with a single attention block and one with two), LSTM, NODE, TCN, and TEMPO-Net.

proportionally, enabling the capture of longer temporal dependencies. This highlights the critical role of RF in EMG-to-kinematics regression, where aligning the model's temporal memory with the decay timescale of the output autocorrelation improved performance. Such alignment allows the model to bridge the mismatch between short-term correlations in the input EMG signals and long-range dependencies in the kinematic outputs [see Fig. 4(b)]. Selecting an RF on the order of several seconds (e.g., 5–10 s or approximately 10 000 samples) is also consistent with the duration of natural motor actions, which supports generalization across individuals with varying movement patterns.

Fig. 4(c) and (d) show predicted versus target finger angles for the two amputee subjects. The model aligns more closely with the target in the accident-related case, with prediction quality resembling that observed in healthy subjects. Fig. 4(e) and (g) display PCA-based visualizations of the ground-truth movements for the two amputees, while Fig. 4(f) and (h) show the corresponding model predictions. The similarity between the predicted and true clusters indicates that the model effectively captures subject-specific movement patterns and underlying temporal structure. This is evident through both visual inspection and the silhouette index of the clustering results. As reported, the silhouette indices for the ground-truth data are 0.48 and 0.58, while the predicted data yield values of 0.40 and 0.58, respectively. Lower silhouette values reflect reduced cluster separability; the consistency between true and predicted indices indicates alignment in cluster structure and supports the reliability of the model's predictions.

#### D. Multitask and Transfer Learning

Fig. 5 illustrates the performance of multitask and transfer learning strategies. In the multitask setting, all subjects are used for pretraining, followed by subject-specific fine-tuning using LoRA or adapters (see Methods). Fig. 5(b) shows the MAE evolution across training iterations for various LoRA

ranks (2, 4, 8, 16, and 24). Higher ranks provide more adaptive capacity, reducing MAE, but also increasing the risk of overfitting. This risk is particularly evident at the highest ranks, where the model may begin to forget shared features learned during pretraining. Fig. 5(a) displays a similar trend for channel-wise adapters, configured to match the trainable parameter count of LoRA. In this formulation, adapters act point-wise at each time step as in the original design. However, a direct comparison of Fig. 5(a) and (b) reveals that this configuration lacks the adaptability of LoRA, as it does not reach equivalent performance minima. To address this, we tested a convolutional adapter variant with reduced channel size (results in Table I), which led instead to pronounced overfitting. These findings highlight that, in data-limited regimes, reusing pretrained nonlinearities via methods like LoRA can be more effective than introducing additional layers.

In the transfer learning setting, average performance across subjects is shown in Fig. 5(d) and (c) for LoRA and adapters, respectively. As in the multitask case, MAE learning curves are reported for various fine-tuning ranks. The results show consistent trends, with higher ranks generally reducing MAE, although overall performance remains lower than in the multitask setting. The above analysis used full data from two sessions for fine-tuning. We now investigate how performance is affected when only limited data is available during adaptation.

Fig. 5(e) shows the effect of data quantity on MAE during fine-tuning in transfer learning, with colors indicating the session from which additional data is drawn (purple: first session; white: second session). MAE decreases progressively with more data, with a marked reduction when data from a second session is added. Fig. 5(f) illustrates the impact of incrementally adding data from alternating sessions. This strategy leads to a rapid decline in MAE, with substantial improvements observed after incorporating only a single gesture from each session.

#### E. Meta-Learning

Fig. 6(a) highlights the conceptual distinction between the two meta-learning schemes. In our setting, where each inner-loop update uses a different minibatch, model-agnostic meta-learning (MAML) and Reptile become nearly equivalent: MAML emphasizes the contribution of the last few inner gradient steps, offering a backward view along the inner-loop trajectory, whereas Reptile aggregates information over the entire trajectory, corresponding to a forward view. Fig. 6(c) shows adaptation performance for a new user using the backward-view strategy (analogous to MAML) under different gradient-combination schemes. Relying only on the final gradient of a trajectory of length 3 ( $g_3$ , light blue) yields suboptimal performance, as it captures little information about the inner optimization path. In contrast, increasing both the number of inner updates and the number of gradients used progressively improves performance, with trajectories of length 20 (red and yellow) outperforming the other settings. These results suggest that exploiting richer information from the inner optimization path leads to lower MAE, moving beyond standard optimization

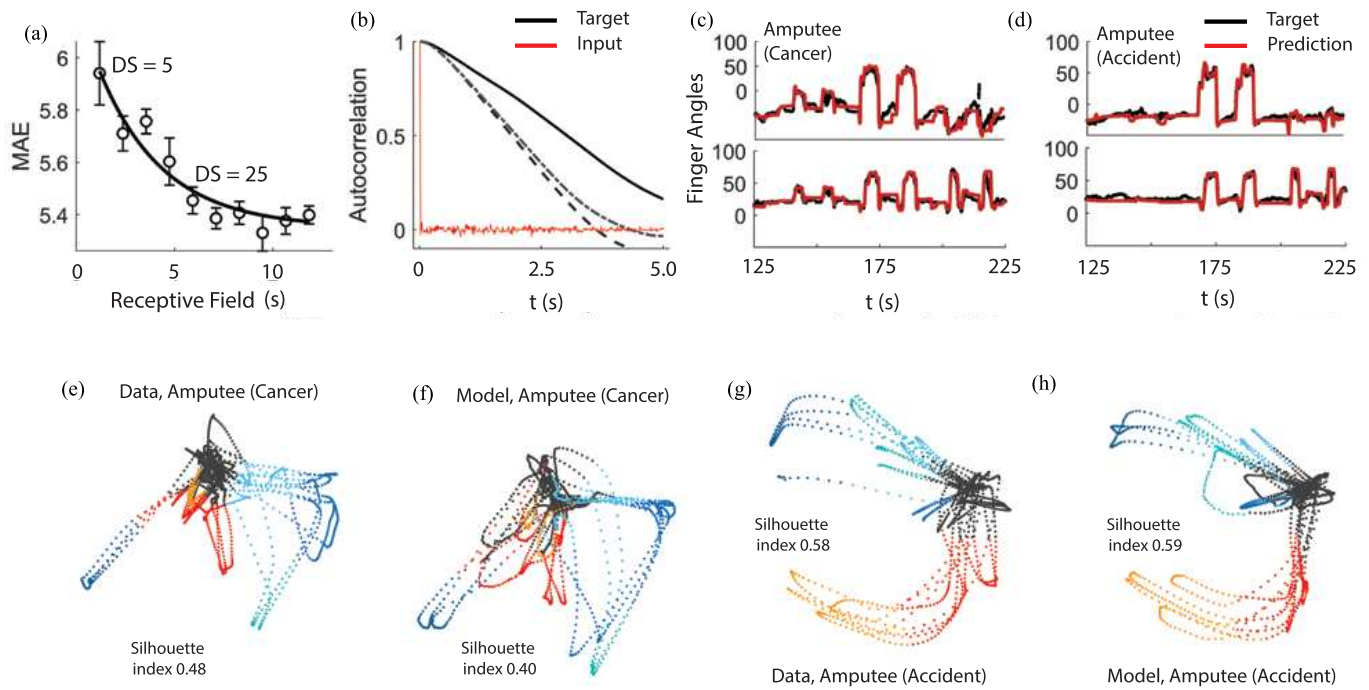


Fig. 4. Overview of results for the TCN model, which consistently achieved state-of-the-art performance. (a) and (b) Impact of DS on RF size and its alignment with the autocorrelation profiles of EMG and target signals. (c) and (d) Predicted versus target finger angles for two amputee subjects, one cancer-related, one accident-related. (e) and (g) present PCA visualizations of ground-truth movement clusters, while (f) and (h) show the corresponding clusters from the model's predictions, highlighting its ability to capture subject-specific temporal structure.

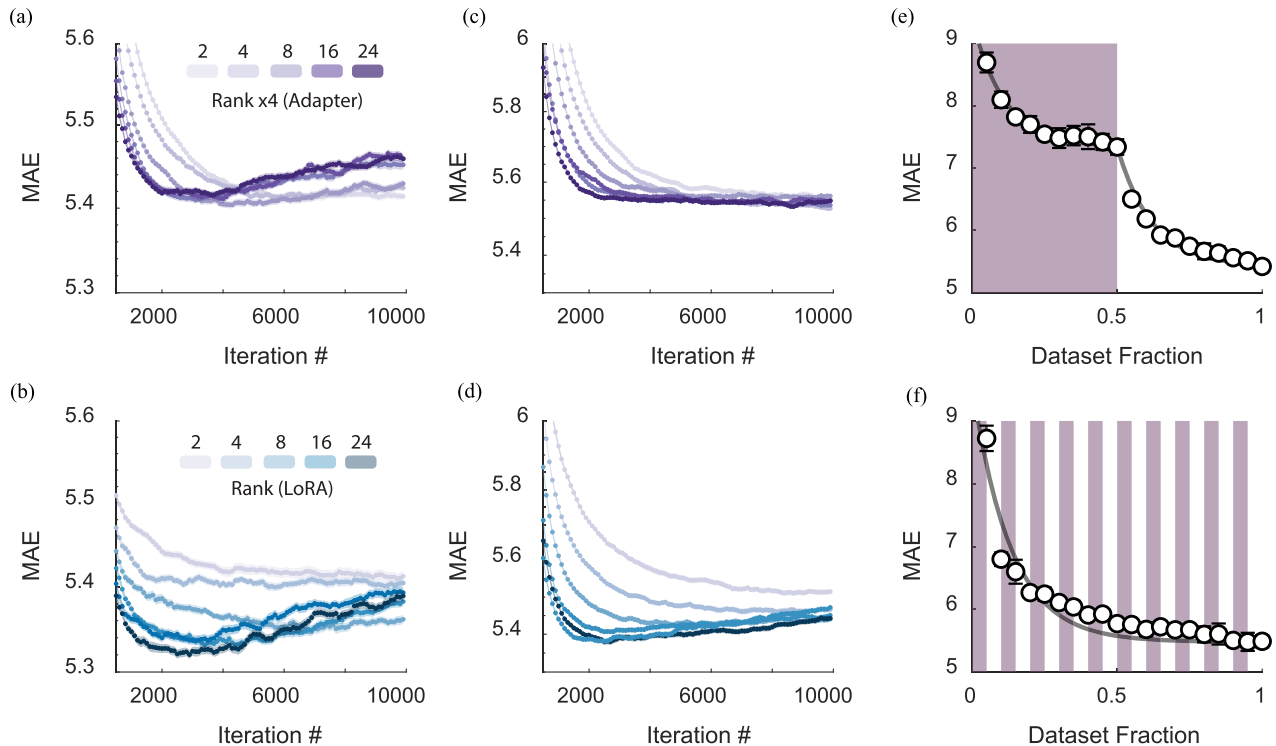
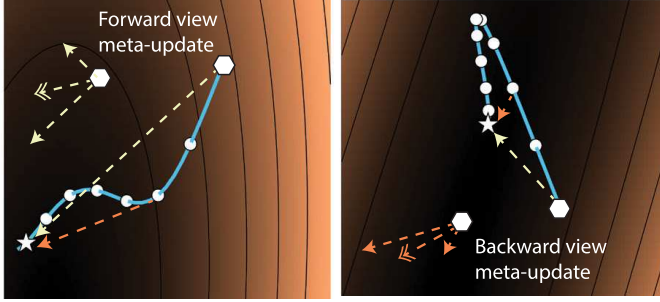


Fig. 5. Overview of multitask and transfer learning approaches. (a) and (b) MAE trends during fine-tuning with adapters and LoRA, respectively, across different parameter matrix ranks, highlighting the effect of fine-tuning dimensionality on performance in multitask learning. (c) and (d) Analogous results for transfer learning, where performance is consistently lower than in the multitask setting. LoRA consistently outperforms adapter layers in both scenarios. (e) and (f) Impact of input data window size on MAE, comparing sequential (Sessions 1 and 2) and alternating session configurations.

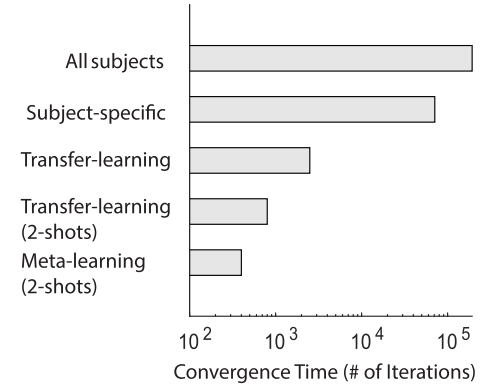
schemes such as transfer learning (dashed line). Furthermore, the improved performance of the yellow curve over the red one indicates that earlier gradients still carry valuable

information for learning shared representations. Such behavior can be expected given the nonmutually exclusive nature of the tasks.

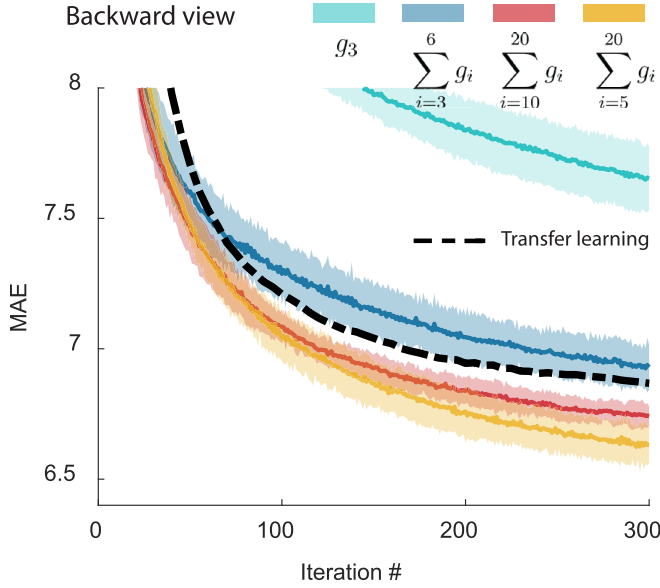
(a) META-learning forward and backward views



(b)



(c)



(d)

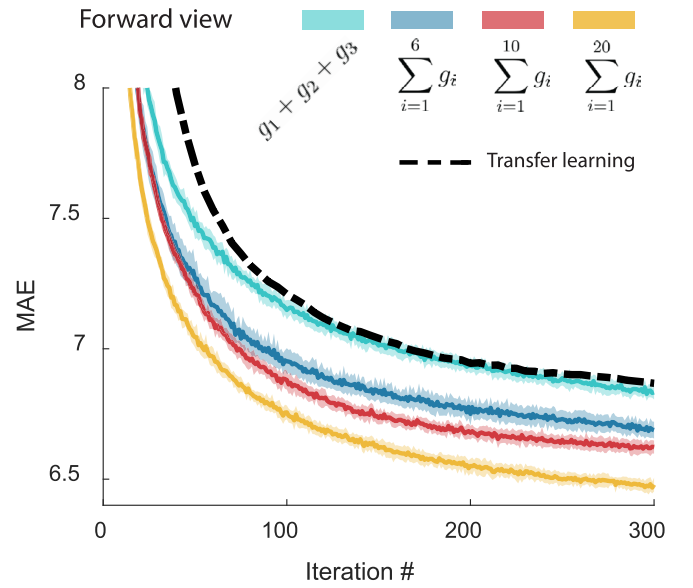


Fig. 6. (a) Schematic comparing forward-view (Reptile-like) and backward-view (MAML-like) meta-updates. Blue trajectories depict inner-loop optimizations for two example tasks over their respective loss surfaces (shown in the two adjacent subpanels). (b) Convergence times for each optimization method. (c) and (d) Test-time adaptation to unseen users for the forward view (c) and backward view (d). Colored curves indicate different ways inner-loop gradients inform the meta-update; dashed lines mark the transfer-learning baseline.

Shifting to the forward view in Fig. 6(d), we observe similar patterns. Increasing the number of gradients used improves task performance, as it enables the model to better capture subject-specific features while allowing the outer loop to focus on learning shared representations. We did not extend the inner-loop trajectories beyond 20 steps, as preliminary experiments indicated diminishing returns for longer sequences relative to their additional computational cost. Comparison of forward- and backward-view results further highlights the importance of early gradients, with the forward view achieving lower MAE. All forward-view configurations consistently outperform the transfer learning baseline, which converges more slowly to a suboptimal minimum and is shown as the dashed lines in the figure. Meta-trained models also exhibit superior one-shot performance, as reported in Table I (few-shot learning section). In this setting, the model is evaluated on unseen subjects and fine-tuned using data from a single session. Meta-learning achieves an MAE of 8.50 with a standard deviation of 1.18, compared to 8.70 and 1.21 for transfer learning. These results confirm that inner and outer loops effectively learn

subject-specific and shared representations, respectively, and support the use of meta-learning for robust HMI systems.

#### F. Overall Comparison and Model Tradeoffs

Table I presents a comprehensive summary of the evaluated models across subject-specific, cross-subject, and adaptive learning settings. Previous baselines from the literature, such as LSTM [15], support vector machine (SVM), MLP [10], Online-Transformer [12], and TempoNet [10], are included for direct comparison on the Ninapro DB8 dataset. Among subject-specific models, the optimized TCN achieves the best performance (MAE = 5.45) with low computational complexity (1.8 millions of multiply-accumulate operations (MMAC), 60k parameters) and an RF of 475 samples (~6 s). The newly introduced Neural-ODE achieves satisfactory accuracy with the fewest parameters (~50k). Across subjects, both TCN and LSTM maintain solid performance with MAE values of 5.70 and 5.57, respectively, while the Online-Transformer shows degraded generalization despite increased model depth. In

adaptive settings, multitask and transfer learning strategies applied to the TCN further improve generalization, achieving MAEs as low as 5.33 (multitask with LoRA). Meta-learning enables fast personalization with minimal data, reaching 6.55 in 2-shot settings.

In addition to performance metrics, Table I also reflects the computational and memory efficiency of each model. Memory usage is approximated from the number of parameters, assuming 32-bit floating point precision (4 B per parameter). For instance, the TCN and NODE models require approximately 240 and 200 KB, respectively, well within the memory budgets of typical embedded microcontrollers.

Overall, the optimized TCN models consistently deliver strong accuracy and efficiency across conditions. The NODE variant offers a promising alternative for low-resource deployment, while adaptive learning methods enable robust generalization in real-world, user-diverse HMI scenarios.

#### IV. DISCUSSION

This work introduces optimized neural network architectures for continuous EMG-to-kinematics regression, establishing a new state-of-the-art across individualized and generalizable models. We adopted the Ninapro DB8 dataset, one of the few public resources providing synchronized high-density EMG and glove-based kinematics from both non-amputee and amputee subjects. Although limited in size and variability, its multimodal structure and wide adoption make it an appropriate benchmark for reproducible comparisons.

##### A. Comparison of Architectures for Single Subjects

We benchmarked five temporal neural architectures, including a windowed self-attention Transformer, to assess whether compact attention mechanisms can capture EMG temporal dynamics efficiently for real-time HMI.

In the subject-specific setting, the optimized TCN achieved the lowest error (MAE = 5.45,  $\sigma$  = 1.06), outperforming TEMPO-Net [10] and the online Transformer [12]. The LSTM and NODE followed closely (MAE = 5.61 and 6.09, respectively, Table I). Fig. 2(a) shows that moderate DS improves performance by extending temporal dependencies, whereas excessive or insufficient RFs degrade generalization by capturing noninformative temporal cues or high-frequency noise.

Because gestures are performed under controlled, repeated conditions (~5 s each), models may learn timing regularities rather than intrinsic EMG features. This limits robustness across sessions, for example, session 3 includes only three gesture repetitions versus ten in sessions 1–2. Random sequence sampling mitigates but does not remove overfitting to long-timescale correlations imposed by the experimental design.

In the across-subject setting, most models exhibited stable performance. The TCN and LSTM achieved comparable accuracy (MAE = 5.70 and 5.57), whereas the NODE showed a notable decline (MAE = 6.86), suggesting limited transferability of its smooth latent dynamics across heterogeneous users.

To address the error accumulation inherent to autoregressive NODEs, we examined hybrid TCN, NODE architectures. Although improvements were marginal, such combinations, using a TCN encoder for preprocessing or numerical solvers in place of residual blocks, may better capture both discrete and continuous temporal dynamics, improving cross-subject generalization.

Overall, the TCN offered the best balance between accuracy, computational cost, and efficiency, making it well-suited for embedded HMI systems. The Transformer consistently underperformed despite lower parameter counts, likely due to its flexible but weak inductive bias, which increases overfitting under limited data. While not inherently unsuitable for prosthetic control, stronger task-specific priors, such as coupling with a TCN encoder or constraining attention hierarchies, may improve stability. Hybrid TCN–Transformer models showed partial gains but remained below standalone TCN performance; larger datasets may better reveal their advantages. Moreover, the Transformer’s structural compatibility with meta-learning frameworks, such as in-context learning, suggests potential for adaptive, low-burden personalization.

Generalizing across heterogeneous users, including amputees, remains a central challenge in neuromuscular decoding. Although the Ninapro DB8 dataset offers synchronized high-density EMG and continuous kinematics, its small cohort and limited variability constrain robustness evaluation. Our analysis shows that meta-learning and multitask learning effectively extend to temporal regression, enabling the meta-trained TCN to adapt rapidly to unseen users with minimal calibration. These findings highlight that robust generalization is attainable through suitable architectural and training choices. Nonetheless, broader validation could be achieved by repurposing classification datasets for regression via inverse kinematics, generating synthetic or physics-based EMG to introduce controlled variability, and leveraging semi- or self-supervised learning to exploit unlabeled data and increase resilience to distribution shifts.

##### B. Fine-Tuning Strategies for Generalizable Models

Training a generic model across all subjects results in a slight performance drop (Fig. 3), indicating the presence of subject-specific features that are challenging to capture with a single parameter set. To assess the potential of neuroprosthetic models to personalize their behavior across users, we investigated multitask learning, transfer learning, and meta-learning.

In this context, multitask learning jointly trains on data from all subjects, capturing both shared and individual patterns. This improves performance while retaining adaptability to unseen users. Transfer learning adopts a two-stage process: pretraining on the full dataset, followed by fine-tuning on a specific subject. This strategy enables rapid adaptation with limited data and consistently outperforms traditional across-subject models and prior baselines (Table I). Transfer learning is particularly valuable in clinical applications, where new users are typically absent from the training set. As shown in Fig. 5(e) and (f), increasing the number of gestures from new sessions improves model accuracy by enriching the statistical



context. Notably, even a single repetition for each gesture from alternate sessions enhances performance, emphasizing the role of temporal diversity in reducing session-specific overfitting and improving robustness. For fine-tuning, we used LoRA and also explored adapter layers as a modular alternative. Adapters incurred slightly higher MAE and computational cost due to the added nonlinear transformations.

Meta-learning is particularly well-suited for HMI applications that require fast adaptation to unseen users. These systems must contend with high intersubject variability and limited data per user. Meta-learning addresses this challenge by training models to generalize from a few samples. In this framework, the inner loop specializes in subject-specific adaptation, while the outer loop learns shared representations across tasks. However, this separation is not guaranteed: the inner loop can converge toward shared structures rather than personalized ones, potentially leading to overfitting. A statistical analysis of patient movement patterns revealed that intersubject variability far exceeds intrasubject variability over time. This confirms that adaptation must address target distribution shifts between users, rather than solely compensate for variations in input conditions. Consequently, some amount of labeled user-specific data remains important. In such scenarios, meta-learning proves more effective than self-supervised test-time adaptation, provided that limited subject data is available.

To tailor meta-learning to this dynamic regression task with noisy EMG inputs, we implemented a custom inner-loop sampling strategy in which mini-batches consisted of temporally downsampled segments from a single subject. We also evaluated FOMAML. Under our sampling scheme, MAML and Reptile are theoretically comparable, and results indicate that leveraging the full inner-loop trajectory during optimization leads to enhanced few-shot learning performance. As shown in Fig. 6, meta-learning achieved lower MAE and faster convergence than traditional transfer learning, a critical advantage in prosthetic applications, where data collection is costly and interuser variability is high. Furthermore, Fig. 6(b) illustrates the convergence time, measured as the number of training iterations required for different learning paradigms to reach stable performance. Meta-learning and transfer-learning in the 2-shot configuration achieve rapid convergence, within just a few hundred iterations, highlighting their efficiency in adapting to new subjects with minimal data. In contrast, standard subject-specific and across-subject training require several orders of magnitude more iterations, reflecting the high computational cost of full retraining.

Table I summarizes the comparative performance across learning strategies. When ample data is available, both multi-task and transfer learning achieve competitive MAE values ( $<5.4$ ), with multitask learning offering a slight edge due to joint optimization across all subjects. These methods are therefore best suited for data-rich contexts. In contrast, meta-learning excels in data-scarce regimes characteristic of practical HMI deployments. In the few-shot setting, meta-trained models consistently outperform transfer learning, with performance further improving as the number of inner-loop steps and updates increases. These results confirm the value of meta-learning in low-data, high-variability settings. While

transfer learning remains effective when more data is available, meta-learning provides faster adaptation, improved robustness, and stronger generalization, critical for scalable, real-time user personalization.

### C. Real-Time Feasibility and Architectural Tradeoffs

A central goal of this study is to evaluate neural network architectures that not only deliver high accuracy, but are also suitable for real-time deployment on embedded or low-power HMI and prosthetic systems. Although EMG signals are typically acquired at high sampling rates (500–2 kHz), this does not necessarily impose excessive computational demands. Inference is generally performed on windowed inputs (e.g., 200 ms to 5 s), and model updates are aligned with the control loop frequency of the assistive device, typically lower than the raw acquisition rate.

For instance, our optimized TCN architecture includes a temporal DS factor of 25, effectively reducing the input rate to approximately 80 Hz, well within the real-time control range (10–50 Hz) relevant for prosthetic applications. This model offers a strong tradeoff between accuracy (MAE = 5.45), parameter count ( $\sim 5.8 \times 10^4$ ), and computational cost (2.0 MMAC). Assuming 32-bit floating-point precision, this results in a memory footprint of approximately 240 KB, compatible with typical embedded microcontroller platforms.

Table I compares the architectural and computational profiles of all evaluated models. The NODE approach, leveraging a continuous-time formulation, produces smooth outputs with the lowest parameter count ( $\sim 4.9 \times 10^4$ ,  $\sim 200$  KB), and remains suitable for real-time use despite its autoregressive nature. The Transformer model, although inherently more computationally intensive due to self-attention, is implemented with a windowed attention mechanism that mitigates resource demands. As a result, it remains practical for deployment ( $\sim 8.9 \times 10^4$  parameters,  $\sim 356$  KB) on low-power systems. Interestingly, the LSTM performs competitively in both accuracy and efficiency when appropriately downsampled and biased toward long temporal dependencies. However, its performance degrades at lower DS rates ( $<50$ ), reflecting its sensitivity to temporal preprocessing and imposing constraints on achievable latency.

Overall, the TCN emerges as the most promising architecture for real-time neuromuscular decoding, offering a compelling balance of accuracy, computational efficiency, and compactness. These characteristics make the TCN particularly well-suited for embedded deployment in low-power HMI applications.

While this work focuses on deep learning-based models for EMG-to-kinematics regression, there is a growing interest in sparse spiking neural networks (SNNs) [10], [44], implemented either as hybrid or fully event-driven systems, for real-time regression. These architectures exploit sparse activity and local learning rules to reduce energy consumption and latency, making them promising for highly constrained embedded environments. Although SNNs currently lags in regression accuracy, recent advances in surrogate gradient optimization, spike-based encoding, and neuromorphic hardware suggest

that they may soon offer viable alternatives for neuromuscular decoding in future applications.

## V. CONCLUSION

This work demonstrates that optimized neural architectures can achieve accurate, real-time EMG-to-kinematics mapping, establishing a new benchmark for HMI. By tuning RFs and applying multitasking, transfer, and meta-learning strategies, deep models effectively capture temporal dependencies critical for continuous finger position estimation.

Intersubject variability can be productively framed as a multitask learning problem, enabling stronger cross-user generalization, a key requirement in prosthetics and teleoperation, where data are scarce, and personalization is vital. Comparative analysis of TCN, Transformer, and NODE models further highlights tradeoffs between accuracy, latency, and memory footprint, supporting embedded deployment.

Progress remains constrained by dataset limitations: the Ninapro DB8 corpus offers limited subject diversity, session variability, and movement speeds, affecting robustness, especially in amputee data. Expanding datasets to include broader conditions would strengthen generalization and inform adaptive receptive-field design, advancing EMG-based control toward reliable, user-adaptive HMI systems.

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